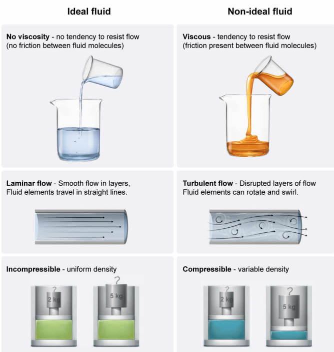
Ideal fluids are used to model the behavior of fluids both in motion and at rest. Which one of the following assumptions does NOT apply to an ideal fluid?

- A. The direction of the flow is equal at all points within a moving fluid. [13%]
- ✔ B. Fluid pressure is not influenced by fluid velocity. [70%]
- C. Frictional forces between fluid molecules are negligible. [4%]
- D. The fluid is incompressible. [11%]

Correct
70% answered correctly

Explanation:



Fluids are substances that have no fixed shape and readily conform to the dimensions of the container in which they are placed. Fluids that are considered **ideal** share the following attributes:

- **No viscosity:** Friction between fluid molecules is negligible such that applied shearing forces (ie, gravity acting on a fluid pouring out of a glass) cause instantaneous, uniform acceleration of the fluid **(Choice C)**.
- **Laminar flow:** The fluid flow is smooth, flowing in parallel layers with no interaction between each layer. In a pipe the fluid elements travel in straight lines and do not swirl around each other **(Choice A)**.
- **Incompressible:** The density of the fluid is modified by neither external forces nor its own weight when oriented in a fluid column **(Choice D)**.

Ideal fluid behavior serves as a model for many practical applications of fluids. For example, water and mercury are both relatively incompressible, which allows both fluids to be used within a simple fluid barometer to reliably measure atmospheric pressure (P_{atm}).

Furthermore, ideal fluid behavior is essential for the application of **Bernoulli's equation**, which mathematically describes the conservation of energy in fluids in terms of fluid pressure (P), density (ρ), and velocity (v) between two points (A and B) within a conduit (ie, the container through which a fluid flows):

$$P_A \qquad\qquad\qquad\qquad + $$

Bernoulli's equation predicts changes in fluid flow and pressure that occur following a change in fluid height (h) or conduit geometry. For example, Bernoulli's equation dictates that the pressure of an ideal fluid will decrease as fluid velocity increases and the height remains constant. Consequently, changes in fluid kinetic energy must be accompanied by changes in pressure energy so that overall energy is conserved.

Therefore, the statement that fluid pressure is not influenced by fluid velocity is not true for an ideal fluid.

Educational objective:
Ideal fluids are totally nonviscous and incompressible; they exhibit smooth, laminar flow without viscosity. Bernoulli's equation dictates that an increase in the velocity of an ideal fluid is accompanied by a decrease in fluid pressure.

Time spent:0 sec
Subject:
Physics

QID:402483
System:
Fluids